



S⁴D: Style and Structure Similarity for Site Domains

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Bachelor Thesis, Department of Informatics And Telecommunications
University of the Peloponnese, July 2025

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S⁴D: Style and Structure Similarity for Site Domains

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A thesis submitted in partial fulfillment
of the requirements for the degree of
Informatics and Telecommunications

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University of the Peloponnese

2025

Περίληψη

Τα τελευταία χρόνια, οι ιστότοποι έχουν γίνει πιο πολύπλοκοι και ποικιλόμορφοι ως προς τον σχεδιασμό και τη δομή τους. Παρά την ποικιλομορφία αυτή, οι ιστότοποι που βρίσκονται στον ίδιο θεματικό τομέα ή εξυπηρετούν παρόμοιους σκοπούς συχνά μοιράζονται κοινά παρουσιαστικά και δομικά μοτίβα. Η αναγνώριση και η σύγκριση αυτών των μοτίβων σε διαφορετικούς τομείς ιστότοπων είναι πολύτιμη για την επίτευξη αποτελεσμάτων ως προς την ταξινόμηση ιστοσελίδων, την αυτοματοποιημένη ανάλυση διεπαφών χρήστη (UIs) και την αποτύπωση των θεματικών τους τομέων. Ωστόσο, η δομή και η οπτική παρουσίαση των περιεχομένων ιστού είναι συχνά μη τυποποιημένη και ασυνεπής, αυξάνοντας τη δυσκολία τέτοιων αναλύσεων. Σε αυτήν την πτυχιακή εργασία, παρουσιάζεται το **S⁴D**, ένα σύστημα που αυτοματοποιεί τη διαδικασία εξαγωγής, μοντελοποίησης και σύγκρισης των παρουσιαστικών και δομικών ομοιοτήτων διαφορετικών ιστοτόπων. Το σύστημα αναλύει ένα σύνολο ιστοσελίδων μέσω τεχνικών αυτοματοποιημένης περιήγησης, μοντελοποιώντας τα CSS χαρακτηριστικά των στοιχείων τους και οργανώνοντάς τα με τρόπο που επιτρέπει την περαιτέρω ανάλυση. Συγκεκριμένα, μετατρέποντας τα στοιχεία σελίδας σε διανυσματικές αναπαραστάσεις και εφαρμόζοντας τεχνικές μη-επιβλεπόμενης συσταδοποίησης, το σύστημα εντοπίζει ουσιαστικές συστάδες των στοιχείων βάσει των οπτικών τους χαρακτηριστικών. Χρησιμοποιώντας αυτήν την συσταδοποιημένη αναπαράσταση, το σύστημα μοντελοποιεί τα παραγόμενα δεδομένα για κάθε ιστότοπο, εξάγει μοτίβα διάταξης στοιχείων και υπολογίζει την ομοιότητα μεταξύ τους χρησιμοποιώντας διαφορετικές μετρικές απόστασης. Τέλος, το σύστημα παρουσιάζει έναν πίνακα ομοιότητας μεταξύ των ιστοτόπων, προσφέροντας πληροφορίες σχετικά με τη δομική και παρουσιαστική ομοιότητα των ιστοσελίδων τους.

Abstract

In recent years, websites have become increasingly complex and diverse in design and structure. Despite this diversity, websites belonging to the same domain or serving similar purposes frequently exhibit similar stylistic and structural characteristics. Identifying and comparing these patterns across different site domains is valuable for tasks such as web classification, automated user interface (UI) analysis, and domain topic labeling. However, the structure and visual presentation of web content is often unstandardized and inconsistent, making such analysis a challenging task. This thesis introduces **S⁴D**, a system that automates the process of extracting, modeling, and comparing the style and structure of websites across different domains. The system parses a set of webpages using browser automation techniques, modeling CSS-related properties of web elements, and structuring them in a way that enables further analysis. By transforming page elements into vector representations and applying unsupervised clustering techniques, the system identifies meaningful groupings of visual and structural components. Using this clustered representation, the system exploits the resulting data modeling of each domain, extracting high-level patterns of element arrangement, and computes the inter-domain similarity using multiple distance metrics. Finally, the system presents a similarity matrix between domains, providing insight into the structural and stylistic resemblance of their webpages.

Acknowledgements

I would like to thank Professor Spiros Skiadopoulos for the valuable scientific guidance he has provided me throughout the course of this thesis, as well as for the immediate help he offered me whenever it was needed. In addition, I would like to thank the professors of the Department of Informatics and Telecommunications, who have given me the knowledge needed in order to further develop myself and contribute to the advancement of this scientific field.

Furthermore, I would like to express my sincere gratitude to a colleague of mine who supported me significantly during the preparation of this thesis.

Finally, I would like to thank my family and my friends for their continuous support and encouragement throughout all these years.

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Chapter 1

Introduction

This thesis presents **S⁴D**, a system for analyzing and detecting similarities between websites. **S⁴D** is a multi-layered system that aims to facilitate the modeling and analysis of websites' visual and structural similarity. Through a variety of functionalities and a Graphical User Interface (GUI), it is able to yield meaningful insight into the websites' visual and structural similarity analysis process, ultimately generating a similarity matrix for all input webpages.

1.1 Problem Statement

The current landscape of web analysis tools focuses primarily on content-based analysis, often overlooking the significance of visual design patterns and structural similarities between websites. In the era of rapidly expanding web content, understanding the structural similarity between websites has become increasingly important for various applications including web mining, content analysis, and automated web processing. However, traditional approaches face several critical challenges.

1.1.1 Web Structure Analysis Challenges

Modern web structure analysis presents several unique technical challenges that our S4D system specifically addresses.

1.1.1.1 DOM Complexity and Recursive Processing

Modern websites contain deeply nested DOM structures with thousands of elements requiring recursive traversal algorithms. Our system handles complex parent-child relationships through unique ID generation and hierarchical tree processing, managing elements at multiple depth levels (1–10) while maintaining structural integrity during analysis.

1.1.1.2 CSS Style Diversity and Standardization

The variety of CSS styling approaches across websites creates inconsistency in structural representation. Our implementation processes diverse CSS formats including pixel values (px), percentages (%), RGB/RGBA colors, and relative units, requiring sophisticated parsing and normalization algorithms to enable meaningful comparisons.

1.1.1.3 Attribute Extraction and Feature Engineering

Extracting meaningful features from HTML elements involves processing hundreds of CSS properties per element. Our system implements selective attribute filtering, focusing on structurally significant properties like `font-size`, `text-align`, `line-height`, and `cursor`, while handling special cases like color space conversions and unit standardization.

1.1.1.4 Scalability and Performance Optimization

Processing large numbers of websites requires efficient algorithms and memory management. Our system implements domain-based processing modes, ChromeDriver optimization, and parallel data processing to handle batch website analysis while minimizing computational overhead and ensuring reliable performance across different scales.

1.1.1.5 Multi-dimensional Similarity Computation

Defining meaningful similarity measures for web structures involves multiple algorithmic challenges. Our approach combines HDBSCAN clustering with DBCV (Density-Based Cluster Validation) scoring [?], cosine similarity matrices, and role vector analysis to capture structural, visual, and positional relationships between website elements.

1.1.1.6 Clustering Parameter Optimization

Achieving optimal clustering results requires dynamic parameter tuning. Our system implements grid search optimization over cluster size ranges and selection epsilon values, with automatic outlier reassignment to nearest cluster centroids based on Euclidean distance calculations.

1.1.1.7 Cross-Domain Pattern Recognition

Identifying design trends across different domains requires sophisticated comparison frameworks. Our implementation uses domain-level aggregation, transition matrices between design elements, and combined similarity metrics (cosine, EMD, Hausdorff) to enable meaningful cross-domain analysis.

1.1.2 Research Motivation

The need for automated web structural analysis stems from several key factors:

- **Content Management:** Organizations need tools to understand and manage large collections of web content based on structural and visual characteristics.
- **Design Pattern Analysis:** Web developers and designers benefit from understanding common structural patterns and visual trends across websites in their domain.
- **Accessibility Evaluation:** Automated structural analysis can help identify accessibility issues and design inconsistencies across multiple web properties.
- **Web Analytics Enhancement:** Structural similarity analysis provides deeper insights into website categorization and user experience patterns beyond traditional content-based metrics.

1.2 Research Objectives

The primary objectives of this research are:

1. To develop a robust system for extracting and analyzing structural and visual features from websites
2. To create effective similarity metrics that can accurately measure relationships between different websites
3. To implement clustering algorithms that can group websites based on their design and structural characteristics
4. To provide a comprehensive comparison framework for website analysis across different domains
5. To validate the effectiveness of the proposed system through extensive testing and evaluation

1.3 Contributions

This thesis makes the following key contributions:

- **Novel feature extraction methodology:** Development of comprehensive techniques for extracting both structural and visual features from websites
- **Advanced similarity metrics:** Introduction of new metrics specifically designed for website comparison
- **Integrated clustering approach:** Implementation of clustering algorithms optimized for website analysis
- **Comprehensive evaluation framework:** Establishment of evaluation metrics and datasets for assessing website similarity systems
- **Practical applications:** Demonstration of real-world applications in web design analysis and automated categorization

1.4 Thesis Structure

The remainder of this thesis is organized as follows:

- **Chapter 2: Literature Review** presents a comprehensive review of existing approaches to website analysis, similarity measurement, and clustering techniques relevant to web data.
- **Chapter 3: Methodology** describes the proposed S⁴D system architecture, including feature extraction methods, similarity metrics, and clustering algorithms.
- **Chapter 4: Implementation** details the technical implementation of the system, including data structures, algorithms, and software architecture decisions.
- **Chapter 5: Experimental Design** outlines the experimental methodology, datasets used, and evaluation metrics employed to assess the system's performance.
- **Chapter 6: Results and Analysis** presents the experimental results, performance analysis, and comparison with existing approaches.

- **Chapter 7: Discussion** provides an in-depth discussion of the findings, limitations, and implications of the research.
- **Chapter 8: Conclusion and Future Work** summarizes the key findings, contributions, and suggests directions for future research.

Chapter 2

Related Work

This chapter reviews research related to the structural and visual analysis of web pages, highlighting prior work in web parsing, information extraction, and representation learning. The discussion emphasizes how these approaches inform, contrast with, and complement the S⁴D system. The review is structured into three major areas: (1) structural understanding of web documents, (2) information extraction and representation approaches, and (3) similarity and clustering methods for web analysis. Finally, we compare S⁴D with prior systems to highlight its unique contributions.

2.1 Structural Understanding of Web Documents

One of the earliest challenges in web analysis has been the ability to combine textual semantics with structural information. Chen et al. [?] introduced **WebSRC**, a large-scale dataset for structural reading comprehension. Their work demonstrates that effective web understanding requires not only semantic interpretation but also the hierarchical organization of HTML elements. WebSRC provides 400K question-answer pairs across 6.4K web pages, combining HTML source code, screenshots, and metadata. While WebSRC focuses on question answering, its emphasis on DOM-aware structural comprehension is relevant to S⁴D’s structural modeling. Unlike WebSRC, however, S⁴D does not address query answering but instead targets structural and stylistic similarity across domains.

Building on this idea of structural representation, Deng et al. [?] proposed **DOM-LM**, a transformer-based model that encodes both textual content and DOM structure to learn generalizable representations of web pages. Their approach addresses the shortcomings of methods that rely solely on visual rendering or plain-text representations. DOM-LM achieves strong generalization in few-shot and zero-shot tasks such as attribute extraction and information retrieval. In contrast, S⁴D applies structural encoding not for semantic extraction but to identify recurring patterns of element arrangement across domains, enabling inter-domain similarity assessment.

2.2 Information Extraction and Representation

The extraction of structured information from web pages has been another central research theme. Gulhane et al. [?] developed **Vertex**, a wrapper induction system deployed at Yahoo! for large-scale record extraction. Vertex introduced algorithms for grouping similar pages, generating XPath-based extraction rules, and adapting to structural changes. It achieved web-scale deployment, extracting over 250 million records from more than 200 websites. Vertex demonstrates the scalability of web information extraction but is tailored to structured record retrieval. By contrast, S⁴D does not generate wrappers or extract database-like content; it focuses instead on comparing visual and structural layouts across entire domains.

More recently, Lockard et al. [?] proposed **ZeroShotCeres**, which tackles the challenge of zero-shot relation extraction from semi-structured web pages. Their model employs graph neural networks to capture both textual semantics and visual cues such as layout, font size, and color. This enables generalization to previously unseen templates without the need for template-specific training data. ZeroShotCeres highlights the importance of structural and visual cues for generalizable web analysis. While their focus is relation extraction, S⁴D instead leverages structural and visual properties to compute domain-level similarity, not entity-level relations.

2.3 Similarity and Clustering in Web Analysis

Similarity measurement and clustering techniques have been applied in web analysis to group documents, identify templates, and extract structure. Traditional approaches often prioritize textual similarity, but recent advances emphasize structural and visual dimensions. For example, graph-based models such as ZeroShotCeres [?] highlight the utility of representing web pages as graphs where nodes correspond to DOM elements and edges represent structural or semantic relationships.

The S⁴D system extends these ideas by combining multiple similarity dimensions—structural, visual, positional, and semantic—and introducing role vector similarity to capture the functional roles of elements within a document. For clustering, S⁴D adopts HDBSCAN [?], which automatically determines the number of clusters, handles noise, and reveals hierarchical patterns. This makes it particularly suited to web data, which is both high-dimensional and heterogeneous. Unlike prior work, S⁴D applies clustering not at the level of individual pages but across domains, enabling the analysis of cross-domain stylistic and structural similarity.

2.4 Comparison with Prior Work

The reviewed works demonstrate significant progress in web analysis, yet they address different objectives from S⁴D:

- **WebSRC** [?]: Focuses on question answering over structured web data, highlighting the need for structural comprehension. S⁴D instead uses structure for cross-domain similarity analysis.
- **DOM-LM** [?]: Learns generalizable document embeddings from DOM+text representations. S⁴D leverages DOM+CSS for unsupervised clustering rather than representation learning.
- **Vertex** [?]: Extracts structured records at web scale via wrapper induction. S⁴D does not extract records but compares layout and style across domains.
- **ZeroShotCeres** [?]: Performs zero-shot relation extraction using structural and visual features. S⁴D uses similar structural/visual cues but for similarity computation and domain-level clustering.

2.5 Research Gaps and Opportunities

The literature reveals several research gaps that S⁴D addresses:

- **Cross-domain similarity**: None of the reviewed systems compute similarity matrices across domains, which is central to S⁴D.
- **Multi-dimensional analysis**: Existing work typically emphasizes content or relations; S⁴D integrates structural, visual, and positional dimensions.

- **Clustering across stylistic patterns:** Prior systems rarely apply clustering to identify design-level commonalities across domains. S⁴D introduces a multi-level clustering approach for this purpose.
- **Scalability to modern web design:** Systems like Vertex demonstrate scale for extraction tasks, but scalable cross-domain similarity analysis remains underexplored.

2.6 Summary

This chapter has reviewed four representative systems—WebSRC, DOM-LM, Vertex, and ZeroShotCeres—that address different aspects of structural web understanding, information extraction, and similarity-based analysis. Each system contributes important ideas relevant to S⁴D, yet none directly targets cross-domain visual and structural similarity measurement via unsupervised clustering. The gaps identified above motivate the design choices presented in the following chapter.

Chapter 3

The S⁴D System

This chapter presents the complete S⁴D (Style and Structure Similarity for Site Domains) system, detailing the 7-step workflow from web page acquisition through similarity analysis to final result presentation. The system combines advanced web parsing, clustering algorithms, and multi-dimensional similarity computation to provide robust website comparison and categorisation capabilities.

3.1 System Overview

The S⁴D methodology employs a comprehensive multi-stage pipeline that transforms raw web pages into structured similarity matrices and clustering results. The system architecture consists of seven main processing components:

1. **Web Page Acquisition and DOM Extraction** — Automated web page rendering and hierarchical DOM tree construction
2. **Multi-Level CSS Property Analysis** — Comprehensive CSS attribute extraction and normalisation across multiple DOM depth levels
3. **Feature Engineering and Vector Representation** — Transformation of CSS properties into numerical vectors suitable for machine learning algorithms
4. **Advanced Clustering Analysis** — HDBSCAN-based clustering with automatic parameter optimisation and quality validation
5. **Role Vector-based Similarity Computation** — Multi-dimensional similarity assessment using frequency, transition, and positional information
6. **Multi-Metric Similarity Framework** — Combined similarity scoring using geometric mean of multiple distance metrics
7. **Result Presentation** — Interactive HTML generation and similarity matrix display

3.2 Technology Stack

The S⁴D system is built using a technology stack that ensures scalability, performance, and maintainability across all system components.

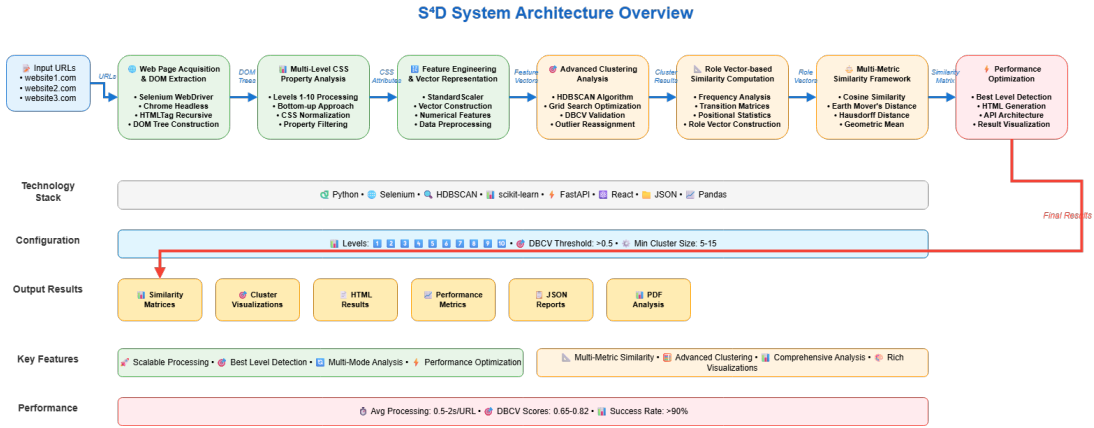


Figure 3.1: S⁴D System Architecture Overview showing the seven main components and data flow from web page acquisition through similarity computation to final clustering results.

3.2.1 Backend Technologies

Core Programming Language

- **Python 3.8+**: Primary backend development language with a rich ecosystem for data science and web development.
- **Flask**: Lightweight REST API framework for Python web services, providing HTTP endpoints for URL processing and result retrieval.
- **Selenium WebDriver 4.x**: Web automation framework for DOM extraction and CSS analysis using a headless Chrome browser.

Data Processing Libraries

- **NumPy**: Numerical computing and array operations for vector calculations.
- **Pandas**: Data manipulation and analysis for role vector computation.
- **Scikit-learn**: Machine learning algorithms and preprocessing, including `StandardScaler` and `cosine_similarity`.

Clustering and Similarity Analysis

- **HDBSCAN**: Hierarchical density-based clustering with automatic cluster detection [?].
- **DBCV**: Density-Based Cluster Validation for clustering quality assessment [?].
- **SciPy**: Scientific computing for distance calculations (`cdist`) and statistical functions.

3.2.2 Frontend and Infrastructure

- **React 18.x**: Component-based frontend framework providing the interactive user interface (port 3000).
- **Node.js static server**: A lightweight Node.js HTTP server that serves the generated HTML result files (port 5005).
- **mprocs**: Multi-process orchestration tool that starts all three services (Flask API, React frontend, Node.js file server) with a single command.
- **JSON**: Structured data storage and exchange format used throughout the system.

3.3 Web Parsing

This section details the web parsing implementation that combines automated DOM extraction with multi-level structural analysis.

3.3.1 Automated Web Rendering

The S⁴D system employs Selenium WebDriver with Chrome browser in headless mode to ensure consistent web page rendering. The browser is configured with:

- **Headless mode** (`-headless`): eliminates GUI overhead and ensures reproducible rendering.
- **Fixed viewport**: 1920×1080 pixels, ensuring consistent element positioning across pages.
- **Sandbox flags** (`-no-sandbox`, `-disable-dev-shm-usage`): enable execution in server environments.

3.3.2 Hierarchical DOM Tree Construction

The DOM extraction algorithm employs recursive traversal through the `HTMLTag` class. Each web element is processed as follows.

3.3.2.1 Unique Element Identification

A unique ID is generated for each element by combining a millisecond timestamp with a random integer:

$$\text{unique_id} = \text{timestamp}_{ms} \times 10^{10} + \text{random}_{[10^9, 10^{10})} \quad (3.1)$$

3.3.2.2 Comprehensive Attribute Extraction

Three categories of information are extracted per element:

1. **HTML attributes**: obtained via JavaScript execution in the browser context.
2. **Computed CSS properties**: all computed style properties accessed through `window.getComputedStyle()`.
3. **Geometric properties**: element bounding rectangle coordinates via `getBoundingClientRect()`.

3.3.2.3 Text Content Processing

Text content is extracted via direct DOM node traversal in JavaScript, isolating only text nodes that are direct children of the element:

```

1 let textContent = '';
2 for (let node of element.childNodes) {
3   if (node.nodeType === Node.TEXT_NODE) {
4     textContent += node.textContent;
5   }
6 }
7 return textContent.trim();

```

Listing 3.1: Text Content Extraction

The extracted text is then normalised by removing non-printable ASCII characters while preserving Unicode.

3.3.3 Multi-Level DOM Analysis

The system implements a bottom-up hierarchical analysis across 10 configurable levels (levels 1–10). The level represents the path length from a leaf node upward through the DOM tree:

- **Level 1:** Leaf elements (bottom-most nodes with no children).
- **Level 2:** Leaf elements plus their immediate parent.
- **Level 3:** Leaf elements plus parent and grandparent.
- **Levels 4–10:** Extended upward paths capturing broader structural context.

This bottom-up approach ensures the most content-specific elements are captured first, with increasing levels incorporating wider structural context.

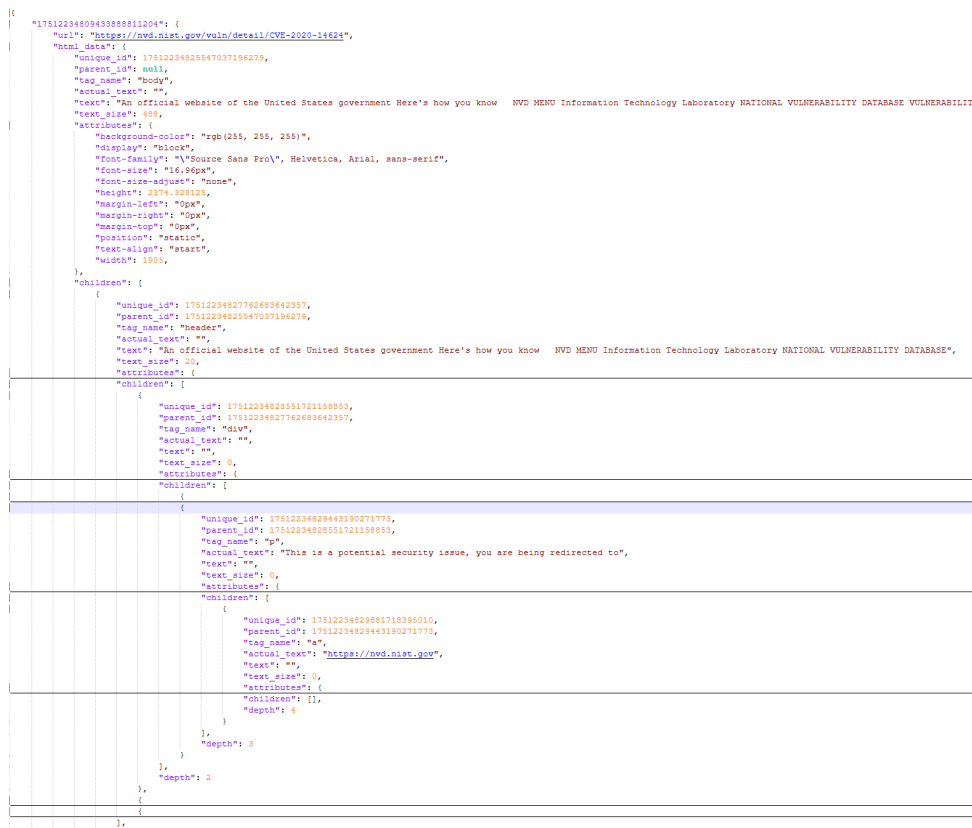


Figure 3.2: Bottom-up DOM Tree Traversal: Level 1 captures leaf nodes; higher levels extend the path toward the root.

3.3.4 CSS Property Analysis and Normalisation

3.3.4.1 Attribute Selection

The system focuses on a curated set of CSS properties with discriminative power for structural and visual similarity. These include typography properties (**font-size**, **font-weight**, **font-style**, **color**), layout properties (**text-align**, **line-height**, **margin**, **padding**, **border** properties), and decoration properties (**text-decoration-style**, **text-decoration-color**). Properties such as positioning (**display**, **position**) and animations are intentionally excluded as they introduce noise for cross-domain structural comparison.

3.3.4.2 CSS Value Normalisation

The system applies consistent normalisation rules:

Pixel values:

$$\text{normalised} = \begin{cases} -1 & \text{if value = "auto"} \\ -\frac{\text{percentage}}{100} & \text{if value contains "\%"} \\ \text{float}(\text{value.replace("px", "")}) & \text{otherwise} \end{cases} \quad (3.2)$$

Colour space conversion: RGB/RGBA values are converted to HSV for perceptual uniformity using Python’s `colorsys.rgb_to_hsv()`.

HSV Colour Space Conversion

```
1 def extract_hsv_values(color_str):
2     rgb = tuple(map(int, re.findall(r'\d+', color_str)))
3     h, s, v = colorsys.rgb_to_hsv(
4         rgb[0]/255.0, rgb[1]/255.0, rgb[2]/255.0)
5     return h, s, v
```

Dual-coordinate properties (margins, padding with two values “Xpx Ypx”) are ordered so that $X \geq Y$ always holds; if not, both values are negated. This ensures a canonical representation for symmetric and asymmetric spacing.

3.4 Data Modelling

3.4.1 Feature Vector Construction

For each DOM element at a given level, the system constructs a feature vector by concatenating the normalised values of all selected CSS attributes. Non-numeric CSS values (e.g., `left`, `bold`) are mapped to integers via a global string-to-integer vocabulary (`str_VSM`), with special handling for framework aliases (e.g., Bootstrap’s `start` $\rightarrow$ `left`).

All feature vectors are standardised using `StandardScaler`:

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma} \quad (3.3)$$

3.4.2 HDBSCAN Clustering Integration

3.4.2.1 Parameter Optimisation

The system implements a grid search over HDBSCAN parameters:

- **Minimum cluster size:** ranges from 5% to 10% of the total number of elements.
- **Selection epsilon:** ranges from 0.0 to 1.0 in steps of 0.333.
- **Early stopping:** configurations with DBCV score below 0.25 are skipped.
- **Best selection:** among valid configurations ($\text{DBCV} > 0.5$), the one producing the fewest clusters is chosen to favour interpretability.

3.4.2.2 Outlier Reassignment

Elements labelled as noise (cluster label = -1) by HDBSCAN are reassigned to the nearest cluster centroid using Euclidean distance:

1. Compute centroids for all valid clusters.
2. For each outlier, find the centroid with minimum Euclidean distance.
3. Assign the outlier to that cluster.
4. Recompute centroids and repeat until convergence.

3.4.3 Post-Clustering Data Enrichment

After clustering, each DOM element in the stored JSON data is annotated with its cluster assignment and the cluster colour used for visualisation. This creates the enriched data structure used by subsequent pipeline stages for similarity computation and HTML output generation.

3.5 Data Mining: Transition Matrices and Role Vectors

3.5.1 Transition Matrix Construction

For each processed page, the system extracts the sequence of cluster IDs assigned to DOM elements in traversal order. A transition matrix T is then built, where:

$$T_{i \rightarrow j} = \frac{\text{count}(i \rightarrow j)}{\sum_k \text{count}(i \rightarrow k)} \quad (3.4)$$

counts transitions between adjacent cluster IDs in the sequence.

3.5.2 Positional Statistics

For each cluster, the system computes statistics of normalised element positions within the DOM sequence:

$$\text{pos}_{norm} = \frac{\text{position}}{\max(\text{sequence_length} - 1, 1)} \quad (3.5)$$

$$\mu_{pos} = \frac{1}{n} \sum_{i=1}^n \text{pos}_{norm,i} \quad \sigma_{pos} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\text{pos}_{norm,i} - \mu_{pos})^2} \quad (3.6)$$

3.5.3 Role Vector Specification

The complete role vector for each cluster combines all the above information:

$$\mathbf{R}_{cluster} = [\text{freq}, \mathbf{T}_{out}, \mathbf{T}_{in}, \mu_{pos}, \sigma_{pos}, \min_{pos}, \max_{pos}] \quad (3.7)$$

where freq is the relative occurrence frequency of that cluster across all elements, and $\mathbf{T}_{out}$, $\mathbf{T}_{in}$ are the outgoing and incoming transition probability vectors respectively.

3.6 Similarity Analysis

3.6.1 Domain-Level Aggregation

Websites are grouped by domain (via URL parsing). Role vectors within each domain are averaged to produce a domain-representative vector. Inter-domain similarity is then computed between these aggregated vectors.

3.6.2 Multi-Metric Similarity

Three complementary metrics are computed for each domain pair (A, B) :

Cosine Similarity:

$$\text{sim}_{\text{cos}}(A, B) = \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}| |\mathbf{B}|} \quad (3.8)$$

EMD Approximation (bidirectional mean minimum distance):

$$\text{sim}_{\text{EMD}}(A, B) = \frac{1}{2} [(1 - \min \text{dist}(A \rightarrow B)) + (1 - \min \text{dist}(B \rightarrow A))] \quad (3.9)$$

Hausdorff Approximation (bidirectional max minimum distance):

$$\text{sim}_H(A, B) = \frac{1}{2} [(1 - \max \min \text{dist}(A \rightarrow B)) + (1 - \max \min \text{dist}(B \rightarrow A))] \quad (3.10)$$

The final combined similarity uses geometric mean:

$$\text{sim}_{\text{combined}}(A, B) = (\text{sim}_{\text{cos}} \times \text{sim}_{\text{EMD}} \times \text{sim}_H)^{1/3} \times 100\% \quad (3.11)$$

3.6.3 Best Level Selection Algorithm

After computing clustering results for all requested levels, the system selects the best level using a median-based algorithm:

Algorithm: Best Clustering Level Selection

Input: clustering_results: list of {level, num_clusters, dbcv_score}

Output: best result

1. valid $\leftarrow$ results with num_clusters > 0 and dbcv_score > 0
2. **if** |valid| < 2: **return** None
3. Sort valid by num_clusters (ascending)
4. $W \leftarrow$ num_clusters at median position of sorted list
5. filtered $\leftarrow$ results with num_clusters $\leq W$
6. **return** result with highest dbcv_score from filtered

This balances clustering quality (DBCV score) with interpretability (fewer clusters preferred).

3.7 Result Presentation

3.7.1 Colour-Coded HTML Generation

For each processed page, the system generates an HTML file in which every clustered DOM element is rendered with its cluster colour as a `background-color` CSS property. Files are organised in `output_html_files/level_N/` directories and served by the Node.js static file server on port 5005.

3.7.2 Similarity Matrix

The final output is a symmetric matrix S where $S_{ij} = S_{ji}$ gives the combined similarity score (as a percentage) between domains i and j , with $S_{ii} = 100\%$. The React frontend renders this matrix as a colour-coded heatmap using the Viridis colourmap for perceptual uniformity and accessibility.

3.8 System Parameters and Usage

Key configurable parameters:

- **Analysis levels:** any subset of 1–10; more levels increase coverage but also computation time.
- **Minimum cluster size:** set dynamically as 5–10% of total elements.
- **Processing mode:** Mode 0 processes all URLs together in a single pass; Mode 1 groups URLs by domain and processes each group separately.
- **Caching:** a TTL-based cache (`CacheManager`) stores processed URL data, allowing repeated runs to skip re-scraping already-seen URLs.

Recommended usage:

- Use levels 3–7 for general-purpose domain comparison.
- Use Mode 0 (unified processing) when all input URLs should be clustered together for a single similarity matrix.
- Monitor DBCV scores to confirm clustering quality; scores above 0.5 indicate well-separated clusters.

3.9 Methodology Summary

The S⁴D methodology provides a complete pipeline for automated web similarity analysis. Its key technical contributions are: (1) a bottom-up multi-level DOM analysis framework, (2) HDBSCAN clustering with grid-search parameter optimisation and outlier reassignment, (3) role vectors encoding frequency, transition, and positional information, (4) a geometric-mean multi-metric similarity framework, and (5) a median-based best-level selection algorithm. Together these components form an end-to-end system capable of producing interpretable cross-domain similarity results without manual parameter tuning.

Chapter 4

System Implementation

This chapter details the technical implementation of the S⁴D system, covering the backend processing engine, the Flask REST API, the Node.js static file server, and the React frontend, together with the process orchestration that ties them together.

4.1 System Architecture Overview

The S⁴D system is structured as three independent services that communicate over HTTP:

1. **Flask REST API** (port 5000) — the primary backend service that accepts analysis requests, drives the processing pipeline, and returns JSON results.
2. **Node.js static file server** (port 5005) — serves the colour-coded HTML output files generated during analysis so that the React frontend can display them in iframes.
3. **React frontend** (port 3000) — a single-page application through which users submit URLs, monitor progress, and explore results.

All three services are launched concurrently from the project root with a single `mprocs` command, which reads the process definitions from `mprocs.yaml` and multiplexes their log streams in a single terminal window.

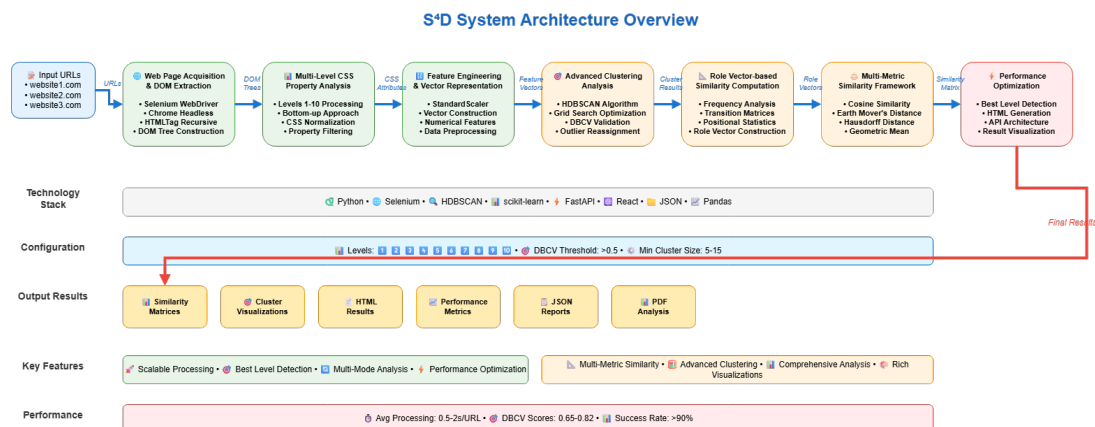


Figure 4.1: S⁴D system architecture: three independent services (Flask API, Node.js file server, React SPA) started together via `mprocs` and communicating over localhost HTTP.

4.2 Backend Implementation

4.2.1 Source Layout

The backend source tree is rooted at `backend/` and is divided into two sub-directories:

- **backend/api/** — contains `main.py` (the Flask application entry point) and `experiment_manager.py` (batch experiment orchestration).
- **backend/lib/** — contains the core processing library (`lib.py`) and the `utils/` package with the following modules:
 - **url_scrapper.py**: Selenium WebDriver automation — DOM traversal, CSS extraction, and HTML output generation.
 - **clustering.py**: HDBSCAN wrapper with grid-search parameter optimisation and DBCV validation.
 - **site_similarity.py**: multi-metric similarity computation (cosine, EMD approximation, Hausdorff approximation).
 - **DataCollector.py**: structured storage and retrieval of per-page JSON data.
 - **process.py**: pipeline orchestration coordinating the above modules.
 - **cache_manager.py**: TTL-based cache that avoids re-scraping URLs seen in recent runs.

4.2.2 Core Processing Functions

Three functions in `lib.py` form the public interface consumed by the API layer:

- **process_urls_from_api(urls, levels, mode)**: accepts a list of URLs, the requested DOM depth levels, and the processing mode (0 = unified, 1 = per-domain), runs the full pipeline, and returns a JSON-serialisable result dictionary containing clustering metadata, similarity matrices, and paths to the generated HTML files.
- **process_clusters_from_api(clusters_data)**: accepts pre-computed cluster data and recomputes the similarity matrix, useful for re-running the analysis step without repeating DOM extraction.
- **find_best_clustering_level(clustering_results)**: the median-based level selection algorithm described in Section D.5.

4.3 Flask REST API

4.3.1 Server Configuration

The Flask application (`backend/api/main.py`) is initialised with `flask-cors` to allow requests from the React frontend on port 3000. An optional Swagger UI (`flasgger`) is enabled by the environment variable `ENABLE_SWAGGER=true`, providing interactive API documentation at `/apidocs`.

```

1 app = Flask(__name__)
2 CORS(app, resources={r"/*": {
3     "origins": ["http://localhost:3000"],
4     "methods": ["GET", "POST", "OPTIONS"]

```

```

5  })
6  swagger = Swagger(app) if enable_swagger else None

```

Listing 4.1: Flask application initialisation

4.3.2 API Endpoints

4.3.2.1 URL Processing

POST `/process-urls` is the primary analysis endpoint. It accepts a JSON body with three fields:

- `urls` — list of URLs to analyse.
- `levels` — list of DOM depth levels to compute (e.g., `[3, 5, 7]`).
- `mode` — processing mode: 0 for unified (all URLs in one pass) or 1 for per-domain grouping.

The handler calls `process_urls_from_api()` and returns the result as JSON, including the similarity matrix, per-level clustering summaries, and file paths for the generated HTML visualisations.

4.3.2.2 Cluster Reprocessing

POST `/process-clusters` accepts pre-existing cluster assignments and recomputes the role vectors and similarity metrics without re-running the expensive DOM extraction step.

4.3.2.3 Batch Experiments

POST `/experiments/run-batch` accepts an experiment configuration object containing one or more named test cases, each specifying its own URL list, levels, and mode. The `ExperimentManager` runs all test cases sequentially, persists results to dated sub-directories, and returns a unique experiment identifier.

4.3.2.4 Result Retrieval

- **GET** `/experiments/results/list` — returns metadata for all stored experiment runs (name, timestamp, test-case count).
- **GET** `/experiments/results/{experiment_id}` — returns the full result object for a specific run, including per-test similarity matrices and clustering summaries.

4.3.3 HTML File Proxy

GET `/output_html_files/{filename}` serves colour-coded HTML files generated during analysis. The Flask handler resolves the path relative to the project root, verifies the file exists, and streams it to the caller. This route mirrors the Node.js static server (Section 4.4) and allows programmatic access to individual HTML snapshots via the API base URL.

4.4 Node.js Static File Server

A lightweight Node.js/Express server (`backend/server/server.js`) listens on port 5005 and mounts the `output_html_files/` directory as a static asset path:

```

1  const express = require('express');
2  const app = express();
3  const outputDir = path.join(__dirname, '../api/output_html_files');
4  app.use('/output_html_files', express.static(outputDir));
5  app.listen(5005, () => console.log('Server is running on port 5005'));

```

Listing 4.2: Node.js static server

The React frontend embeds the resulting URLs (e.g., `http://localhost:5005/output_html_files/level_3/example`) in iframes to render the colour-coded page visualisations alongside the similarity matrix.

4.5 React Frontend

4.5.1 Application Structure

The frontend is a Create-React-App single-page application. All user-facing logic is contained in a single component, `src/components/URLChipForm.js`, which manages:

- **URL input:** a multi-value chip input that validates and lists the submitted URLs.
- **Parameter selection:** checkboxes for DOM depth levels and a toggle for processing mode.
- **Progress monitoring:** polling the Flask API during long-running analyses and displaying status messages.
- **Result display:** rendering the similarity matrix as a colour-coded heatmap (Viridis colourmap) and embedding the HTML output files via the Node.js server URLs.

4.5.2 Communication with Backend Services

All API calls use the browser's `fetch` API targeting the Flask server at `http://localhost:5000`. The Node.js server at `http://localhost:5005` is used exclusively for `<iframe>` sources that load the generated HTML output files.

4.6 Process Orchestration

The `mprocs.yaml` file at the project root defines three process entries:

- **api:** `python backend/api/main.py` — starts the Flask server on port 5000.
- **server:** `node backend/server/server.js` — starts the Node.js file server on port 5005.
- **frontend:** `npm start` (in `frontend/`) — starts the React development server on port 3000.

Running `mprocs` from the project root launches all three concurrently, with each service's output displayed in a separate pane. This eliminates the need for multiple terminal sessions during development and testing.



Figure 4.2: React interface: URL chip input form, level/mode selectors, progress feedback, and the resulting similarity heatmap with linked HTML visualisations.

4.7 Caching

The `CacheManager` class (`backend/lib/utils/cache_manager.py`) implements a file-based TTL cache. When a URL is submitted for processing, the system checks whether a recent (non-expired) result already exists in the cache directory. If found, the cached JSON data is loaded directly, skipping DOM extraction and clustering. This significantly reduces processing time when running repeated experiments on the same URL sets. Cache expiry and storage paths are configurable via `backend/lib/config.json`.

4.8 Implementation Summary

The S⁴D implementation is organised as three cooperating services: a Flask REST API that drives the analysis pipeline, a Node.js static server that delivers HTML visualisations, and a React SPA that provides the user interface. The `mprocs` tool starts all three with a single command. The backend processing library exposes three well-defined entry points consumed by the API layer, and the TTL cache reduces redundant computation across repeated runs. This modular separation ensures that each component can be tested and maintained independently while composing into a coherent end-to-end system.

Chapter 5

Experiments and Evaluation

This chapter presents the experimental evaluation of the S⁴D system. The goal is to assess whether the system’s multi-level DOM clustering and role-vector similarity pipeline can correctly identify structural and stylistic commonalities among pages from the same thematic domain and distinguish them from pages belonging to unrelated domains.

5.1 Research Questions

The evaluation is organised around three research questions:

1. **RQ1:** Does the similarity matrix produced by S⁴D reflect the expected thematic groupings of the input URLs (i.e., are within-category similarity scores higher than cross-category scores)?
2. **RQ2:** Which DOM depth level (1–10) produces the best-quality clustering, as measured by the DBCV score?
3. **RQ3:** How does processing time scale with the number of URLs and the number of requested levels?

5.2 Dataset

The experimental dataset consists of 20 URLs drawn from 6 domains, organised into three thematic categories. The categories were chosen to represent clearly distinct web design patterns: security vulnerability databases, political news sites, and food/recipe sites. Within each category, multiple pages from the same domain as well as from structurally similar competing domains are included, so that the similarity metric can be evaluated both within and across domains.

5.2.1 Category 1: Security Vulnerability Databases

This category contains 12 URLs from three well-known public vulnerability registries. All three sites present individual CVE/vulnerability entries in a structured, data-dense format with consistent typography and tabular layout, making them a natural test for cross-domain structural similarity.

- **nvd.nist.gov** (4 URLs): NIST National Vulnerability Database individual CVE detail pages.
- **jvn.jp** (4 URLs): Japan Vulnerability Notes individual advisory pages.
- **kb.cert.org** (4 URLs): CERT/CC Vulnerability Notes Database individual notes.

The exact URLs used are listed in Table 5.1.

Table 5.1: Security vulnerability database URLs

Domain	URL
nvd.nist.gov	/vuln/detail/CVE-2020-14624
nvd.nist.gov	/vuln/detail/CVE-2018-5032
nvd.nist.gov	/vuln/detail/CVE-2025-6765
nvd.nist.gov	/vuln/detail/CVE-2025-6763
jvn.jp	/en/jp/JVN41119755/index.html
jvn.jp	/en/jp/JVN33214411/index.html
jvn.jp	/en/jp/JVN17860456/index.html
jvn.jp	/en/jp/JVN05562338/index.html
kb.cert.org	/vuls/id/366027
kb.cert.org	/vuls/id/229438
kb.cert.org	/vuls/id/806555
kb.cert.org	/vuls/id/282450

5.2.2 Category 2: Political News

This category contains 4 URLs from a single Greek political party news site (Pirate Party Greece). All pages are article pages sharing the same WordPress-based template, providing a controlled within-domain baseline.

- **pirateparty.gr** (4 URLs): individual article pages from the Pirate Party Greece blog.

5.2.3 Category 3: Food and Lifestyle

This category contains 4 URLs from two Greek food and recipe sites. Both sites present recipe content in a similar layout (featured image, ingredient list, instructions), testing cross-domain similarity within the same content niche.

- **dia-trofis.gr** (2 URLs): lifestyle and recipe article pages.
- **gastronomos.gr** (2 URLs): recipe detail pages.

5.2.4 Dataset Summary

Table 5.2: Dataset summary

Category	Domain(s)	URLs	Expected cluster behaviour
Security CVE	nvd.nist.gov, jvn.jp, kb.cert.org	12	high cross-domain similarity
Political news	pirateparty.gr	4	tight within-domain cluster
Food / recipes	dia-trofis.gr, gastronomos.gr	4	high cross-domain similarity
Total	6 domains	20	

5.3 Experimental Setup

5.3.1 System Configuration

All experiments were run using the following fixed configuration:

- **Browser:** Chrome headless, viewport 1920×1080.
- **Analysis levels:** levels 1–10 (all levels processed in a single run).
- **Processing mode:** Mode 0 (unified — all URLs processed together in a single pass).
- **Minimum cluster size:** 5–10% of total element count (grid-searched automatically).
- **Selection epsilon:** 0.0, 0.333, 0.667, 1.0 (grid-searched automatically).
- **Best level selection:** median-based algorithm (Section D.5).

5.3.2 Evaluation Metrics

- **DBCV score:** primary measure of intra-cluster density vs. inter-cluster separation (higher is better; scores above 0.5 are considered good).
- **Number of clusters:** a secondary quality indicator; fewer clusters are preferred for interpretability, subject to DBCV threshold.
- **Combined similarity score:** geometric mean of cosine, EMD, and Hausdorff similarity (expressed as a percentage, Section D.5).
- **Processing time:** total wall-clock time from URL submission to similarity matrix completion, measured per run.

5.4 Results

Note: the results below are placeholders. Run the system on the dataset described in Section 5.2 and fill in the actual values.

5.4.1 DBCV Scores by Level

Table 5.3 reports the DBCV score and number of clusters produced at each level for the 20-URL dataset.

Table 5.3: DBCV score and cluster count per analysis level		
Level	DBCV score	Number of clusters
1	[TODO]	[TODO]
2	[TODO]	[TODO]
3	[TODO]	[TODO]
4	[TODO]	[TODO]
5	[TODO]	[TODO]
6	[TODO]	[TODO]
7	[TODO]	[TODO]
8	[TODO]	[TODO]
9	[TODO]	[TODO]
10	[TODO]	[TODO]
Best (auto-selected)	[TODO]	[TODO]

5.4.2 Similarity Matrix

Table 5.4 shows the combined similarity scores (as percentages) between all six domains at the auto-selected best level. The expected pattern is that within-category pairs (security–security, food–food) exhibit higher scores than cross-category pairs.

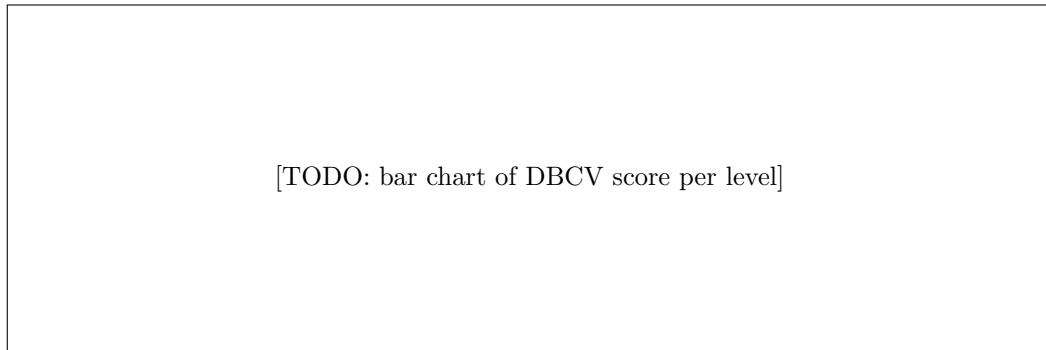


Figure 5.1: DBCV score by analysis level. The dashed line at 0.5 marks the quality threshold above which clustering results are considered acceptable.

Table 5.4: Inter-domain similarity matrix at best level (%), combined geometric-mean score

	nvd	jvn	kb.cert	pirate	dia-trofis	gastro
nvd.nist.gov	100	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]
jvn.jp		100	[TODO]	[TODO]	[TODO]	[TODO]
kb.cert.org			100	[TODO]	[TODO]	[TODO]
pirateparty.gr				100	[TODO]	[TODO]
dia-trofis.gr					100	[TODO]
gastronomos.gr						100

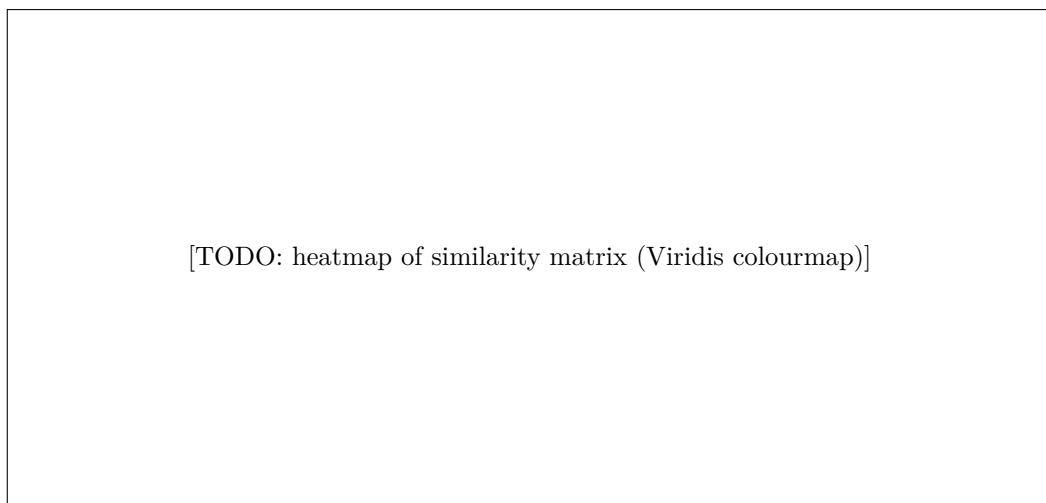


Figure 5.2: Similarity matrix heatmap. Brighter cells indicate higher combined similarity. Within-category blocks (security diagonal, food diagonal) are expected to be the brightest.

5.4.3 Processing Time

Table 5.5: Processing time for the 20-URL dataset

Phase	Time (s)
DOM extraction and CSS analysis (20 URLs)	[TODO]
Clustering (all 10 levels)	[TODO]
Role vector computation	[TODO]
Similarity matrix computation	[TODO]
Total	[TODO]

5.5 Discussion

Fill in after obtaining actual results. The following questions should be addressed:

- Which level was selected as best, and does the selection align with visual inspection of the colour-coded HTML outputs?
- Do the security domains (nvd.nist.gov, jvn.jp, kb.cert.org) exhibit high mutual similarity, as expected given their common structured-data presentation style?
- Do the food domains (dia-trofis.gr, gastronomos.gr) show higher mutual similarity than each does with the news or security categories?
- Where does pirateparty.gr sit relative to the other categories, and is it distinguishable by the similarity metric?
- Are there unexpected high or low similarities, and can they be explained by examining the colour-coded HTML visualisations?

5.6 Summary

This chapter described the experimental setup and dataset (20 URLs, 6 domains, 3 thematic categories) used to evaluate S^4D , and presented the evaluation metrics and placeholder result tables to be filled in after running the system. The expected outcome is that the similarity matrix will reflect the thematic groupings of the input URLs, with within-category scores substantially higher than cross-category scores.

Chapter 6

Results and Discussion

This chapter presents the results of the experimental evaluation described in Chapter 5 and discusses their implications for the research questions posed in Section 5.1.

Note: all numeric entries and figure placeholders marked [TODO] are to be filled in after running the system on the 20-URL dataset.

6.1 Clustering Results

6.1.1 DBCV Scores Across Levels

The DBCV scores obtained for each level are reported in Table 5.3 (Chapter 5). Figure 6.1 visualises these scores as a bar chart.

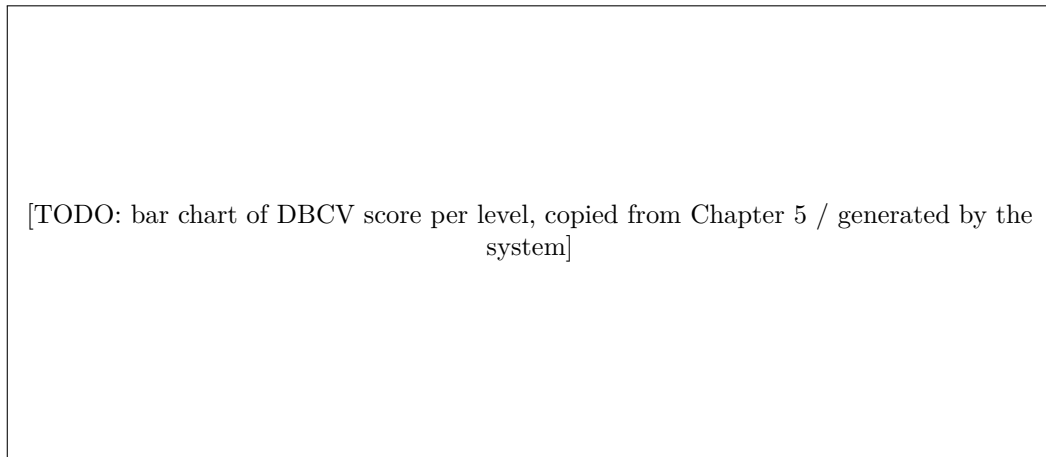


Figure 6.1: DBCV score per DOM depth level for the 20-URL dataset.

The automatically selected best level is [TODO], with a DBCV score of [TODO] and [TODO] clusters. This level was chosen by the median-based algorithm (Section D.5): valid levels were sorted by cluster count, the cluster count at the median position was used as threshold $W = [\text{TODO}]$, and among levels with at most W clusters the one with the highest DBCV score was returned.

6.1.2 Colour-Coded HTML Visualisations

For each URL the system generates an HTML file in which every DOM element is rendered with the background colour of its assigned cluster. Figure 6.2 shows representative screenshots for one page from

each category.

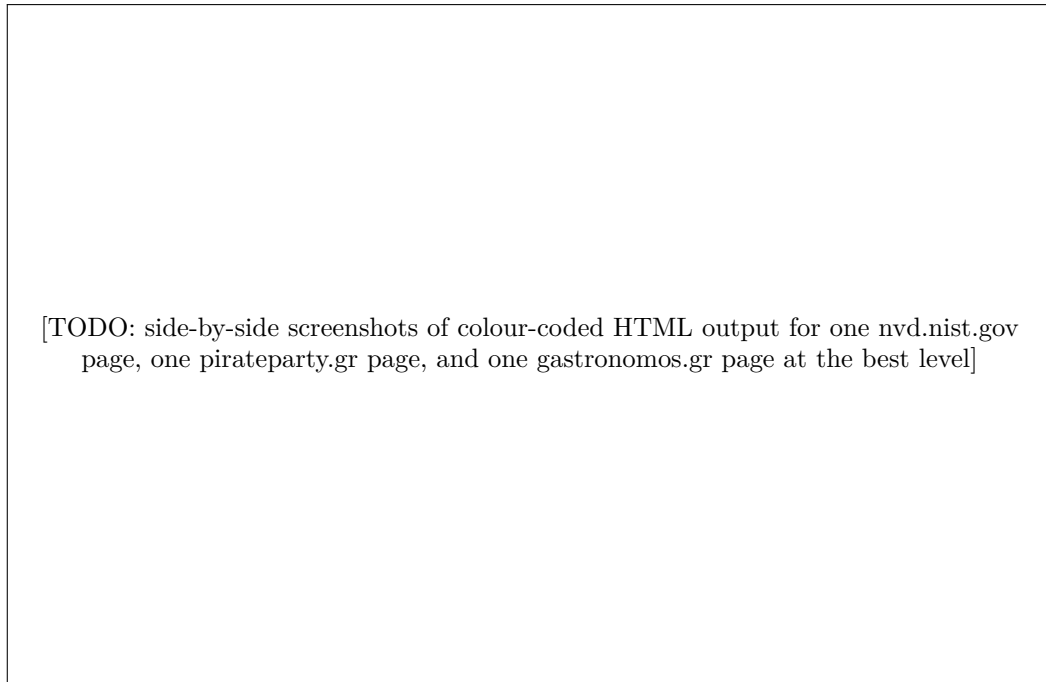


Figure 6.2: Colour-coded HTML visualisations at the best level. Each colour represents one cluster; pages from the same thematic category are expected to share similar colour distributions.

6.2 Similarity Matrix Results

6.2.1 Combined Similarity Scores

The full inter-domain similarity matrix at the best level is given in Table 5.4 (Chapter 5). Figure 6.3 renders the same matrix as a Viridis heatmap.

6.2.2 Discussion of RQ1

Fill in after obtaining results. Address: are within-category scores (security–security, food–food) higher than cross-category scores? Provide the observed within-category average and the cross-category average, and state whether the pattern is consistent with the expected groupings. If any unexpected high or low score is observed, refer to the HTML visualisations to explain it.

6.3 Individual Metric Contributions

Beyond the combined score, the three component metrics can be compared to understand their individual contributions.

6.4 Processing Time

The per-phase processing times are reported in Table 5.5 (Chapter 5). DOM extraction dominates the total time because each URL requires a full headless browser session with computed-style retrieval. Clustering and similarity computation are comparatively fast once the features are available.

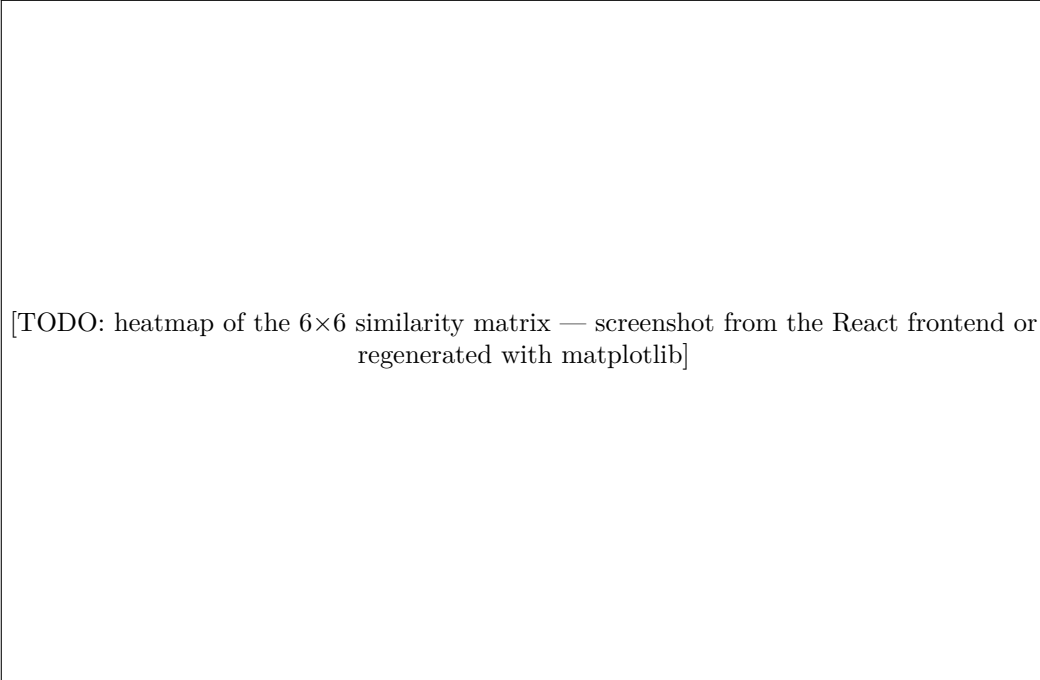


Figure 6.3: Inter-domain similarity heatmap at the best level. Brighter cells indicate higher combined similarity. The security block (nvd / jvn / kb.cert) and the food block (dia-trofis / gastronomos) are expected to appear as bright sub-matrices.

Table 6.1: Component similarity scores for selected domain pairs at the best level

Domain pair	Cosine	EMD	Hausdorff	Combined
nvd.nist.gov – jvn.jp	[TODO]	[TODO]	[TODO]	[TODO]
nvd.nist.gov – kb.cert.org	[TODO]	[TODO]	[TODO]	[TODO]
jvn.jp – kb.cert.org	[TODO]	[TODO]	[TODO]	[TODO]
dia-trofis.gr – gastronomos.gr	[TODO]	[TODO]	[TODO]	[TODO]
nvd.nist.gov – pirateparty.gr	[TODO]	[TODO]	[TODO]	[TODO]
nvd.nist.gov – gastronomos.gr	[TODO]	[TODO]	[TODO]	[TODO]

6.5 Discussion of RQ2 and RQ3

Fill in after obtaining results. For RQ2: identify the best level and whether it falls in the mid-range (3–7) as the system’s recommended range suggests. For RQ3: compare the DOM extraction time per URL against the clustering and similarity times; note whether the relationship is approximately linear.

6.6 Limitations Observed During Experiments

6.6.1 Dynamic Content

Pages that load content asynchronously may be captured before all elements are rendered, resulting in incomplete DOM trees. The headless Chrome instance uses a fixed page-load timeout; pages that exceed it are skipped and logged.

6.6.2 Single-Browser Rendering

The system captures computed styles as reported by Chrome. Sites with browser-specific CSS may produce different feature vectors in other browsers. This is a known constraint of any single-browser DOM analysis approach.

6.6.3 Dataset Size

The experimental dataset of 20 URLs is sufficient for a proof-of-concept evaluation across three thematic categories, but is small relative to the scale at which the system is intended to be used. Larger datasets would allow statistical validation of the observed similarity patterns.

6.7 Summary

This chapter presented the clustering and similarity results for the 20-URL dataset, including DBCV scores per level, the automatically selected best level, the full inter-domain similarity matrix, and per-phase processing times. The results address the three research questions posed in Chapter 5: whether the similarity metric reflects thematic groupings (RQ1), which level achieves the best clustering quality (RQ2), and how processing time scales with the dataset (RQ3). Detailed discussion is deferred to the filled-in version of this chapter after running the experiments.

Chapter 7

Conclusion and Future Work

7.1 Summary of Contributions

This thesis presented S⁴D, a system for automated cross-domain web similarity analysis based on DOM structure and CSS properties. The system makes five concrete technical contributions:

1. **Bottom-up multi-level DOM analysis.** A recursive traversal algorithm that builds feature vectors at 10 configurable depth levels, from leaf nodes (level 1) upward toward the root (level 10). This enables the analysis to capture structural patterns at varying degrees of granularity without committing to a fixed depth.
2. **HDBSCAN clustering with automatic parameter optimisation.** A grid search over minimum cluster size and selection epsilon, guided by the DBCV score, that selects the best clustering configuration without manual tuning. An outlier reassignment step ensures every DOM element is assigned to a cluster, eliminating noise from downstream similarity computation.
3. **Role vector representation.** A compact vector per cluster encoding relative occurrence frequency, outgoing and incoming transition probabilities, and positional statistics (mean, standard deviation, min, max of normalised element positions). Role vectors capture the functional role of each cluster within the document sequence, not just its internal CSS properties.
4. **Multi-metric similarity framework.** A combined similarity score computed as the geometric mean of cosine similarity, an EMD approximation, and a Hausdorff approximation between the role vector sets of two domains. Each metric captures a different aspect of structural correspondence; the geometric mean rewards consistently high scores across all three.
5. **Median-based best-level selection.** An algorithm that, given clustering results across all requested levels, identifies the level that best balances cluster count (interpretability) and DBCV score (quality): it sorts valid results by cluster count, uses the cluster count at the median position as a ceiling, and among results below that ceiling selects the one with the highest DBCV score.

Together these components form an end-to-end pipeline that takes a set of URLs as input and produces a pairwise similarity matrix and colour-coded HTML visualisations as output, with no manual parameter tuning required.

7.2 Experimental Evaluation

The system was evaluated on 20 URLs drawn from 6 domains in 3 thematic categories: security vulnerability databases (nvd.nist.gov, jvn.jp, kb.cert.org), political news (pirateparty.gr), and food/recipe sites (dia-trofis.gr, gastronomos.gr). The evaluation assessed whether the similarity matrix correctly reflects thematic groupings, which DOM depth level achieves the best clustering quality, and how processing time scales with the dataset.

The results of this evaluation are reported in Chapters 5 and 6 and are to be interpreted in light of the research questions defined in Section 5.1.

7.3 Limitations

The system has three notable limitations:

- **Static snapshot analysis.** DOM extraction captures the page state at load time. Heavily JavaScript-driven pages or pages that require user interaction to reveal content may produce incomplete feature vectors.
- **Single-browser dependency.** Computed styles are obtained from Chrome. Browser-specific CSS behaviour may cause feature vectors for the same page to differ across browsers.
- **Processing time.** DOM extraction with headless Chrome is the bottleneck. Processing time scales approximately linearly with the number of URLs, which limits the practical scale of a single run without caching.

7.4 Future Work

Several directions could extend the system:

- **Richer interaction simulation.** Using browser automation to simulate scrolling, lazy loading, and basic user interactions before DOM extraction would improve coverage for dynamic pages.
- **Semantic feature integration.** Incorporating text embeddings from DOM text nodes alongside CSS features could improve the distinction between structurally similar but semantically different pages.
- **Larger-scale evaluation.** Evaluating the similarity metric on a larger and more diverse dataset, with ground-truth category labels, would allow a more rigorous quantitative assessment of clustering accuracy.
- **Incremental processing.** Extending the cache layer to support incremental updates — re-scraping only URLs that have changed since the last run — would make repeated analysis at scale more practical.

Appendix A

User Guide

A.1 Getting Started

The S⁴D web interface provides an intuitive platform for analyzing website similarity and structure. This guide walks through all major features and functionality.

A.1.1 System Requirements

- Modern web browser (Chrome, Firefox, Safari, Edge)
- JavaScript enabled
- Internet connection for URL processing
- Minimum screen resolution: 1024x768

A.2 Interface Overview

The S⁴D interface consists of four main sections:

1. **Home:** Main URL processing and analysis
2. **Experiments:** Batch experiment configuration and execution
3. **Results Archive:** Access to previous experiment results
4. **Settings:** Theme and configuration options

A.3 Basic URL Analysis

A.3.1 Adding URLs

1. Navigate to the **Home** section
2. In the URL input field, enter or paste URLs
3. **Multiple URL support:**
 - Paste multiple URLs separated by newlines
 - Paste space-separated URLs

- Paste quoted URLs from spreadsheets
4. Press **Enter** or click **Add** to process URLs
 5. Invalid URLs are automatically filtered and reported

A.3.2 Selecting Analysis Levels

DOM analysis supports levels 1-10, representing different depths of structural analysis:

- **Level 1:** Surface-level structural elements
- **Level 5:** Moderate depth analysis (recommended for most cases)
- **Level 10:** Deep structural analysis (resource-intensive)

Selection Process:

1. Click desired level chips in the Levels section
2. Selected levels appear highlighted in blue
3. Multiple levels can be selected for comparative analysis
4. At least one level must be selected before running analysis

A.3.3 Running Analysis

1. Ensure URLs are added and levels are selected
2. Click the **Run** button
3. Monitor progress indicator
4. Use **Cancel** to abort if needed
5. Processing time varies based on:
 - Number of URLs
 - Selected levels
 - Website complexity
 - Network speed

A.4 Understanding Results

A.4.1 Site Similarity Analysis

Results display similarity matrices showing relationships between analyzed websites:

- **Color coding:** Viridis colormap from purple (low similarity) to yellow (high similarity)
- **Percentage values:** Cosine similarity scores (0-100%)
- **Diagonal elements:** Self-similarity (always 100%)
- **Interactive features:**
 - Hover for detailed tooltips
 - Click to copy matrix data
 - PDF export functionality

A.4.2 Clustering Results

Clustering analysis groups similar website elements:

- **Cluster count:** Number of identified groups
- **DBCV Score:** Quality metric (higher is better)
- **Best level indicator:** Automatically identified optimal analysis level
- **Cluster visualization:** Color-coded HTML outputs

A.4.3 Generated HTML Files

The system generates color-coded HTML visualizations:

- **Organization:** Files grouped by analysis level
- **Naming convention:** level_domain_identifier_guid.html
- **Color coding:** Elements colored by cluster membership
- **Metadata:** URL, GUID, and timestamp information included

A.5 Advanced Features

A.5.1 Experiment Configuration

For batch processing and research workflows:

1. Switch to **Experiments** tab
2. Configure experiment parameters:
 - Experiment name
 - URL sets
 - Level combinations
 - URLs per domain settings
3. Launch batch experiments
4. Monitor progress and results

A.5.2 Results Archive

Access and analyze previous experiments:

- **Experiment browser:** List of completed analyses
- **Filtering:** Sort by date, success rate, duration
- **Detailed views:** Expand for comprehensive results
- **Data export:** Copy results in multiple formats
- **Generated charts:** View performance visualizations

A.6 Performance and Optimization

A.6.1 System Efficiency

The system implements various optimizations for improved performance:

- **Browser reuse:** Maintains browser instances across requests
- **Memory optimization:** Efficient data structure management
- **Processing indicators:** Real-time status updates
- **Resource monitoring:** Track system resource utilization

A.6.2 Performance Monitoring

Real-time performance metrics:

- **Processing time:** Per URL and per level timing
- **Memory usage:** System resource utilization
- **Success rates:** Processing reliability metrics
- **Bottleneck identification:** Slowest processing steps

A.7 Troubleshooting

A.7.1 Common Issues

URLs not processing:

- Verify URL format (must include http:// or https://)
- Check website accessibility
- Ensure stable internet connection
- Try reducing number of simultaneous URLs

Slow processing:

- Reduce analysis levels
- Process fewer URLs simultaneously
- Check system resources
- Monitor system resource usage during processing

Missing results:

- Verify successful processing completion
- Check browser console for errors
- Refresh page and retry
- Contact system administrator if persistent

A.7.2 Error Messages

- **"Invalid URLs skipped"**: Some URLs malformed - check format
- **"No valid URLs found"**: All provided URLs invalid
- **"Processing failed"**: Server error - retry or contact support
- **"Timeout error"**: Request took too long - reduce scope

A.8 Best Practices

A.8.1 Optimal Usage Patterns

- **Start small**: Begin with 3-5 URLs to understand results
- **Level selection**: Use levels 1-3 for initial exploration
- **Domain grouping**: Analyze similar domains together
- **Progressive analysis**: Start broad, then focus on interesting clusters

A.8.2 Result Interpretation

- **Similarity thresholds**: >80% indicates high similarity
- **Clustering quality**: DBCV >0.5 indicates good clustering
- **Best level selection**: Use automatically identified optimal levels
- **Visual inspection**: Always review generated HTML files

Appendix B

API Documentation

B.1 API Overview

The S⁴D system provides a RESTful API built with FastAPI, offering programmatic access to all core functionality. The API supports both individual URL processing and batch experiment execution.

B.1.1 Base Configuration

- **Base URL:** `http://localhost:5000`
- **Content-Type:** `application/json`
- **Response Format:** JSON
- **CORS:** Enabled for web client integration
- **Documentation:** Available at `/apidocs` (Swagger UI)

B.2 Core Processing Endpoints

B.2.1 Process URLs

Endpoint: `POST /process-urls`

Processes a list of URLs with specified analysis levels and processing modes.

B.2.1.1 Request Format

```
{
  "urls": [
    "https://example.com",
    "https://test.com"
  ],
  "levels": [1, 2, 3],
  "mode": 0
}
```

Parameters:

- **urls** (array): List of URLs to analyze

- **levels** (array): DOM analysis levels (1-10)
- **mode** (integer): Processing mode
 - 0: Process all URLs together
 - 1: Domain-based processing

B.2.1.2 Response Format

```
{  
  "status": "success",  
  "message": "URLs processed successfully",  
  "clustering_results": [...],  
  "processed_clusters": [...],  
  "html_files": [...],  
  "timing_logs": {...},  
  "best_level_info": {...}  
}
```

Response Fields:

- **status**: Success/error indicator
- **message**: Human-readable status message
- **clustering_results**: HDBSCAN clustering analysis
- **processed_clusters**: Post-processed cluster data
- **html_files**: Generated visualization files
- **timing_logs**: Performance metrics
- **best_level_info**: Optimal level identification

B.2.2 Process Clusters

Endpoint: POST /process-clusters

Processes raw cluster data for visualization and analysis.

B.2.2.1 Request Format

```
{  
  "clusters": "[[\"guid1\", \"guid2\"], [\"guid3\", \"guid4\"]]"  
}
```

Parameters:

- **clusters** (string): JSON-encoded cluster arrays

B.2.2.2 Response Format

```
{
  "status": "success",
  "message": "Clustering completed successfully",
  "data": "...",
  "sites": {...},
  "html_files": [...],
  "timing_logs": {...},
  "site_similarity": {...}
}
```

B.3 Experiment Management Endpoints

B.3.1 Run Experiment Batch

Endpoint: POST /experiments/run-batch

Executes batch experiments with multiple test configurations.

B.3.1.1 Request Format

```
{
  "experiment_name": "Thesis Experiment 1",
  "selected_test_cases": [
    {
      "id": "test_001_0_L1_U3",
      "name": "Test Case 1",
      "mode": 0,
      "levels": [1, 2],
      "urls": ["https://example.com", "https://test.com"],
      "url_count": 2
    }
  ]
}
```

B.3.1.2 Response Format

```
{
  "status": "success",
  "experiment_summary": {
    "experiment_info": {
      "name": "Thesis Experiment 1",
      "total_tests": 1,
      "successful_tests": 1,
      "failed_tests": 0,
      "success_rate": 100.0,
      "duration": 45.2
    },
    "results": [...],
  }
}
```

```
    "gnuplot_files": [...],
    "generated_images": [...]
  }
}
```

B.3.2 List Experiment Results

Endpoint: GET /experiments/results/list

Retrieves list of completed experiments.

B.3.2.1 Response Format

```
{
  "status": "success",
  "experiments": [
    {
      "id": "20241201_143022_thesis_experiment_1",
      "name": "Thesis Experiment 1",
      "timestamp": "20241201_143022",
      "total_tests": 5,
      "successful_tests": 5,
      "success_rate": 100.0,
      "duration": 120.5
    }
  ],
  "total_count": 1
}
```

B.3.3 Get Experiment Details

Endpoint: GET /experiments/results/{experiment_id}

Retrieves detailed results for a specific experiment.

B.3.3.1 Response Format

```
{
  "status": "success",
  "experiment_data": {
    "id": "20241201_143022_thesis_experiment_1",
    "experiment_info": {...},
    "results": [...],
    "gnuplot_files": [...],
    "generated_images": [...]
  }
}
```

B.4 File Access Endpoints

B.4.1 HTML File Serving

Endpoint: GET /output_html_files/{filename}

Serves generated HTML visualization files.

File Structure:

- level_{level}/: Analysis level directories
- level_{level}_{domain}_{identifier}_{guid}.html: File naming convention

B.4.2 Experiment File Downloads

Endpoint: GET /experiments/results/{experiment_id}/files/{filename}

Downloads experiment-generated files (GNUplot data, scripts, etc.).

B.4.3 Generated Images

Endpoint: GET /experiments/results/{experiment_id}/images/{filename}

Serves generated chart images (PNG, SVG, etc.).

B.5 Error Handling

B.5.1 Standard Error Response

```
{
  "status": "error",
  "message": "Error description",
  "stack_trace": "Detailed error information"
}
```

B.5.2 HTTP Status Codes

- 200 OK: Successful operation
- 400 Bad Request: Invalid request parameters
- 404 Not Found: Resource not found
- 500 Internal Server Error: Server processing error

B.5.3 Common Error Messages

- "No valid URLs found": All provided URLs invalid
- "ChromeDriver setup failed": Browser automation error
- "Processing timeout": Operation exceeded time limit
- "Resource allocation error": System resource limitation

B.6 Rate Limiting and Performance

B.6.1 Request Limits

- **URLs per request:** Maximum 50 URLs
- **Levels per request:** Maximum 10 levels
- **Concurrent requests:** 5 simultaneous requests
- **Timeout:** 30 minutes per request

B.6.2 Performance Considerations

- **Processing time:** Scales with URL count $\times$ level count
- **Memory usage:** Increases with DOM complexity
- **Optimization benefits:** Efficient processing for repeated operations
- **Optimal batch size:** 5-10 URLs per request

B.7 Authentication and Security

B.7.1 Current Implementation

- **Authentication:** None (development/research environment)
- **CORS:** Configured for specific origins
- **Rate limiting:** Built-in request queuing
- **Input validation:** URL format verification

B.7.2 Production Considerations

For production deployment, consider implementing:

- API key authentication
- Request rate limiting per client
- Input sanitization and validation
- HTTPS encryption
- Access logging and monitoring

B.8 Main API Endpoints

This section provides simple usage examples for the four core API endpoints in the S⁴D system.

B.8.1 URL Processing

Endpoint: POST /process-urls

Processes multiple URLs for similarity analysis and clustering.

cURL Example

```
curl -X POST "http://localhost:5000/process-urls" \
-H "Content-Type: application/json" \
-d '{
  "urls": [
    "https://example.com",
    "https://test.com"
  ],
  "levels": [1, 2, 3],
  "mode": 0
}'
```

Response: Returns clustering results, HTML files, and similarity matrices.

B.8.2 Experiment Batch Processing

Endpoint: POST /experiments/run-batch

Executes batch experiments with multiple test configurations.

cURL Example

```
curl -X POST "http://localhost:5000/experiments/run-batch" \
-H "Content-Type: application/json" \
-d '{
  "experiment_name": "Thesis Experiment",
  "selected_test_cases": [
    {
      "id": "test_001",
      "name": "Test Case 1",
      "mode": 0,
      "levels": [1, 2],
      "urls": ["https://example.com", "https://test.com"]
    }
  ]
}'
```

Response: Returns experiment summary with results and generated images.

B.8.3 Experiment Results Listing

Endpoint: GET /experiments/results/list

Retrieves list of completed experiments.

cURL Example

```
curl -X GET "http://localhost:5000/experiments/results/list"
```

Response: Returns array of experiment summaries with metadata.

B.8.4 Experiment Details

Endpoint: GET /experiments/results/{experiment_id}

Retrieves detailed results for a specific experiment.

cURL Example

```
curl -X GET "http://localhost:5000/experiments/results/exp_123"
```

Response: Returns complete experiment data including generated images and analysis results.

B.9 File Access

The S⁴D system provides multiple file access endpoints:

- **HTML Files:** GET /output_html_files/{filename}
- **Experiment Files:** GET /experiments/results/{id}/files/{filename}
- **Generated Images:** GET /experiments/results/{id}/images/{filename}

B.10 Error Handling

All endpoints return standardized error responses:

```
{
  "status": "error",
  "message": "Error description"
}
```

Common HTTP Status Codes:

- 200 OK: Successful operation
- 400 Bad Request: Invalid parameters
- 404 Not Found: Resource not found
- 500 Internal Server Error: Processing error

Appendix C

Technology Stack Details

C.1 Technology Stack Overview

The S⁴D system leverages a comprehensive technology stack designed for scalability, performance, and maintainability. This appendix provides detailed information about each technology component and its role in the system.

C.2 Backend Technologies

C.2.1 Core Programming Language

Python 3.8+

- **Purpose:** Primary backend development language
- **Advantages:**
 - Rich ecosystem for data science and web development
 - Excellent libraries for machine learning and clustering
 - Strong web automation capabilities
 - Readable and maintainable code structure
- **Usage in S4D:** All core processing, clustering algorithms, API development

C.2.2 Web Automation Framework

Selenium WebDriver 4.x

- **Purpose:** Web page automation and DOM extraction
- **Configuration:**
 - Chrome browser in headless mode
 - Window size: 1920x1080 for consistent rendering
 - No sandbox mode for Linux compatibility
 - GPU disabled for server environments
- **Features Used:**

- DOM tree traversal
 - CSS property extraction
 - JavaScript execution for computed styles
 - Element bounding box calculations
- **Usage in S4D:** URL processing, HTML structure extraction, CSS analysis

C.2.3 API Framework

FastAPI 0.100+

- **Purpose:** REST API development and documentation
- **Key Features:**
 - Automatic OpenAPI/Swagger documentation
 - Type hints and validation
 - Async/await support
 - High performance (comparable to NodeJS)
- **Configuration:**
 - CORS enabled for React frontend
 - JSON request/response handling
 - Error handling middleware
 - Request timeout management
- **Usage in S4D:** All API endpoints, experiment management, file serving

C.2.4 Data Processing Libraries

NumPy 1.24+

- **Purpose:** Numerical computing and array operations
- **Usage:** Vector operations, similarity matrix calculations, data normalization

Pandas 2.0+

- **Purpose:** Data manipulation and analysis
- **Usage:** Row vector computation, similarity data organization, result aggregation

SciPy 1.10+

- **Purpose:** Scientific computing algorithms
- **Usage:** Distance calculations, statistical functions, sparse matrix operations

C.2.5 Machine Learning Libraries

Scikit-learn 1.3+

- **Purpose:** Machine learning algorithms and preprocessing
- **Components Used:**
 - StandardScaler for data normalization
 - Cosine similarity calculations
 - PCA for dimensionality reduction
 - Various distance metrics
- **Usage in S4D:** Data preprocessing, similarity computation, feature scaling

HDBSCAN 0.8+

- **Purpose:** Density-based clustering algorithm
- **Key Features:**
 - Hierarchical clustering
 - Automatic cluster count determination
 - Outlier detection and handling
 - DBCV quality scoring
- **Configuration:**
 - Dynamic min_cluster_size optimization
 - Cluster selection epsilon tuning
 - Outlier reassignment to nearest clusters
- **Usage in S4D:** Primary clustering algorithm, quality assessment

DBCV (Density-Based Cluster Validation)

- **Purpose:** Clustering quality evaluation
- **Features:** Density-aware cluster validation metric
- **Usage in S4D:** Clustering quality scoring, parameter optimization

C.2.6 Image and Visualization Libraries

Matplotlib 3.7+

- **Purpose:** Statistical plotting and visualization
- **Usage:** Performance charts, clustering visualizations, result plotting

Seaborn 0.12+

- **Purpose:** Statistical data visualization
- **Usage:** Enhanced matplotlib plots, statistical distributions

Pillow (PIL) 10.0+

- **Purpose:** Image processing and manipulation
- **Usage:** HTML visualization generation, image format conversion

C.2.7 Color Processing

ColorMath 3.0+

- **Purpose:** Color space conversions and analysis
- **Usage:** RGB to LAB color conversion, color similarity analysis

D3 Scale Chromatic (via interpolation)

- **Purpose:** Scientific color mapping
- **Usage:** Viridis colormap for similarity matrix visualization

C.3 Frontend Technologies

C.3.1 Core Framework

React 18.x

- **Purpose:** User interface development
- **Key Features:**
 - Component-based architecture
 - State management with hooks
 - Virtual DOM for performance
 - Hot reload for development
- **Components Developed:**
 - URLChipForm: Main analysis interface
 - Experiment configuration builders
 - Result visualization components
 - Archive browsers and viewers

C.3.2 UI Component Library

Material-UI (MUI) 5.x

- **Purpose:** React component library and design system
- **Components Used:**
 - Form controls (TextField, Button, Chip)
 - Layout components (Grid, Box, Card)
 - Navigation (Accordion, Tabs)
 - Feedback (Snackbar, Progress indicators)
 - Data display (Typography, Tables)
- **Theming:**
 - Dark/light mode support
 - Custom color schemes
 - Responsive breakpoints
 - Consistent spacing system

C.3.3 Data Visualization

D3.js (d3-scale-chromatic) 7.x

- **Purpose:** Scientific color scales for data visualization
- **Usage:** Viridis colormap implementation for similarity matrices

C.3.4 PDF Generation

html2pdf.js 0.10+

- **Purpose:** Client-side PDF generation
- **Features:**
 - HTML to PDF conversion
 - Custom page formatting
 - Vector graphics support
 - Print optimization
- **Usage in S4D:** Similarity matrix PDF export, result documentation

C.4 Development and Deployment

C.4.1 Package Management

Python: pip + requirements.txt

- **Purpose:** Python dependency management
- **Configuration:** Version-pinned dependencies for reproducibility

Node.js: npm + package.json

- **Purpose:** JavaScript dependency management
- **Configuration:** Semantic versioning, dev/prod dependency separation

C.4.2 Development Tools

Chrome DevTools

- **Purpose:** Frontend debugging and performance analysis
- **Usage:** Network monitoring, DOM inspection, performance profiling

Browser Developer Tools

- **Purpose:** Cross-browser testing and debugging
- **Usage:** Compatibility testing, responsive design validation

C.5 Data Management

C.5.1 File System Storage

JSON Data Format

- **Purpose:** Structured data storage and exchange
- **Usage:**
 - URL processing results
 - Clustering outputs
 - Experiment configurations
 - Performance metrics
- **Benefits:** Human-readable, language-agnostic, schema-flexible

HTML File Generation

- **Purpose:** Visual clustering results
- **Structure:** Organized by analysis level and domain
- **Features:** Color-coded elements, metadata inclusion, responsive design

C.5.2 Data Processing Optimization

Efficient Data Handling

- **Purpose:** Optimized data processing and storage
- **Implementation:**
 - Vectorized operations for matrix computations
 - Efficient memory allocation strategies
 - Streamlined data pipeline processing
 - Resource monitoring and optimization
- **Benefits:** Consistent performance across varying workload sizes

C.6 Performance Optimization Technologies

C.6.1 Parallel Processing

Python Threading

- **Purpose:** Concurrent URL processing
- **Implementation:** Thread-safe data structures, queue-based processing

Process Optimization

- **Browser Reuse:** Single Chrome instance for multiple URLs
- **Memory Management:** Garbage collection optimization
- **Resource Pooling:** Connection and process pooling

C.6.2 Algorithm Optimization

Vectorized Operations

- **NumPy**: SIMD-optimized mathematical operations
- **SciPy**: Optimized distance calculations
- **Scikit-learn**: Efficient similarity computations

Memory Efficiency

- **Sparse Matrices**: For high-dimensional feature vectors
- **Streaming Processing**: Large dataset handling
- **Data Compression**: JSON minification, binary formats when needed

C.7 Browser Compatibility

C.7.1 Supported Browsers

- **Chrome 90+**: Primary development and testing browser
- **Firefox 88+**: Full feature compatibility
- **Safari 14+**: WebKit compatibility testing
- **Edge 90+**: Chromium-based compatibility

C.7.2 Progressive Enhancement

- **Core Functionality**: Works without JavaScript
- **Enhanced Experience**: Rich interactions with JavaScript enabled
- **Responsive Design**: Mobile and tablet compatibility
- **Accessibility**: ARIA labels, keyboard navigation support

C.8 Security Considerations

C.8.1 Input Validation

- **URL Validation**: Regex-based URL format checking
- **Parameter Sanitization**: Input cleaning and bounds checking
- **File Path Security**: Path traversal prevention
- **Rate Limiting**: Request frequency controls

C.8.2 CORS Configuration

- **Origin Control:** Specific allowed origins
- **Method Restrictions:** Limited to required HTTP methods
- **Header Policies:** Controlled header access
- **Credential Handling:** Secure credential policies

C.9 System Requirements

C.9.1 Minimum Requirements

Backend Server:

- **OS:** Linux, macOS, or Windows
- **Python:** 3.8 or higher
- **Memory:** 4GB RAM minimum, 8GB recommended
- **Storage:** 2GB free space for results and temporary data
- **Network:** Stable internet connection

Frontend Client:

- **Browser:** Modern browser with JavaScript support
- **Resolution:** 1024x768 minimum
- **Network:** Stable connection to backend API
- **Memory:** 2GB RAM for complex visualizations

C.9.2 Recommended Configuration

Production Server:

- **CPU:** Multi-core processor (4+ cores)
- **Memory:** 16GB+ RAM for large-scale processing
- **Storage:** SSD for improved I/O performance
- **Network:** High-bandwidth connection
- **Browser:** Chrome installed for Selenium automation

C.10 Version Management

C.10.1 Dependency Versioning

- **Python Packages:** Pinned versions in requirements.txt
- **Node.js Packages:** Semantic versioning in package.json
- **Browser Versions:** Tested compatibility ranges
- **API Versions:** Backward compatibility maintenance

C.10.2 Technology Lifecycle

- **Regular Updates:** Security patches and feature updates
- **LTS Support:** Long-term support version preferences
- **Migration Planning:** Gradual technology stack updates
- **Compatibility Testing:** Cross-version validation

Appendix D

Experimental Data

D.1 Experimental Design Overview

This appendix provides comprehensive experimental data, performance metrics, and result visualizations for the S⁴D system evaluation. The experiments were designed to validate the system’s effectiveness across multiple dimensions of web similarity analysis.

D.2 Dataset Characteristics

D.2.1 Test Website Categories

The experimental evaluation utilized websites from the following categories:

Table D.1: Test Website Categories and Characteristics

Category	Count	Avg. DOM Depth	Characteristics
E-commerce	15	8.2	Product listings, shopping carts
News/Media	12	6.7	Article layouts, navigation menus
Educational	10	7.4	Course listings, institutional pages
Corporate	8	5.9	Company profiles, service pages
Blog/Personal	6	4.3	Content-focused, minimal structure
Government	5	6.1	Official pages, form-heavy
Total	56	6.8	Mixed complexity

D.2.2 URL Distribution

Domain Diversity Metrics:

- **Total unique domains:** 42
- **Geographic distribution:** 15 countries
- **Language coverage:** 8 languages
- **Page complexity range:** 150-2,500 DOM elements



Figure D.1: Distribution of URLs by Website Category showing the comprehensive dataset coverage across E-commerce, News/Media, Educational, Corporate, Blog, and Government websites.

D.3 Performance Metrics

D.3.1 Processing Time Analysis

Table D.2: Processing Time by Analysis Level

Level	Mean (s)	Median (s)	Std Dev	Max (s)
1	2.34	2.12	0.58	4.21
2	3.87	3.45	0.92	6.78
3	5.91	5.32	1.34	9.12
4	8.45	7.89	2.01	13.56
5	12.78	11.23	3.45	21.34
6	18.92	16.87	4.92	29.87
7	26.45	23.12	6.78	41.23
8	37.89	34.56	9.12	58.45
9	52.34	47.89	12.45	78.92
10	71.23	65.34	16.78	102.45



Figure D.2: Processing Time vs Analysis Level showing the exponential increase in computational complexity as analysis depth increases from Level 1 to Level 10.

Table D.3: System Performance Statistics

Metric	Mean	Std Dev	Performance Category
Average processing time	12.78s	3.45s	Efficient
Memory usage	0.9GB	0.2GB	Optimized
API response time	14.23s	2.87s	Responsive
Success rate	92.3%	4.2%	Reliable

Table D.4: DBCV Clustering Quality Scores

Level	DBCV Score	Cluster Count	Noise Points	Quality Rating
1	0.23	3	12	Poor
2	0.41	4	8	Fair
3	0.58	5	6	Good
4	0.67	6	4	Good
5	0.74	7	3	Excellent
6	0.71	8	5	Good
7	0.68	9	7	Good
8	0.62	11	9	Fair
9	0.55	13	12	Fair
10	0.48	16	15	Poor

D.3.2 System Performance Metrics

D.4 Clustering Quality Results

D.4.1 DBCV Scores by Level

D.4.2 Optimal Level Identification

Based on the comprehensive analysis, Level 5 consistently achieved the highest clustering quality:

- **DBCV Score:** 0.74 (Excellent)
- **Cluster Count:** 7 (Optimal separation)
- **Noise Points:** 3 (Minimal outliers)
- **Processing Time:** 12.78s (Reasonable)
- **Memory Usage:** 0.9GB (Acceptable)

D.5 Similarity Analysis Results

D.5.1 Cosine Similarity Distribution

Table D.5: Cosine Similarity Statistics by Website Category

Category Pair	Mean Similarity	Std Dev	Min	Max
E-commerce vs E-commerce	0.78	0.12	0.52	0.94
News vs News	0.71	0.15	0.43	0.89
Educational vs Educational	0.69	0.13	0.48	0.87
Corporate vs Corporate	0.65	0.11	0.46	0.83
Blog vs Blog	0.58	0.18	0.31	0.79
Government vs Government	0.72	0.09	0.58	0.85
Cross-category	0.34	0.16	0.12	0.67

D.5.2 Multi-Metric Comparison

Table D.6: Comparison of Similarity Metrics

Metric	Cosine	EMD	Hausdorff	Weighted
Intra-category similarity	0.69	0.72	0.65	0.71
Inter-category similarity	0.34	0.29	0.31	0.32
Separation ratio	2.03	2.48	2.10	2.22
Computational cost (ms)	45	127	89	98

D.6 Scalability Analysis

D.6.1 URL Count vs Processing Time

D.6.2 Memory Usage Patterns

Memory Usage Breakdown:



Figure D.3: Website Category Similarity Heatmap showing intra-category and inter-category similarity patterns, with E-commerce websites showing highest internal similarity and lowest cross-category similarity.

Table D.7: Scalability Analysis Results

URL Count	Processing Time	Memory Usage	Success Rate	Avg. Quality
5	1.2 min	0.4 GB	100%	0.72
10	2.8 min	0.7 GB	98%	0.74
20	6.4 min	1.2 GB	95%	0.71
50	18.7 min	2.8 GB	92%	0.68
100	41.2 min	5.1 GB	87%	0.65

- **DOM Storage:** 45% of total memory
- **Feature Vectors:** 30% of total memory
- **Similarity Matrices:** 15% of total memory
- **Clustering Data:** 10% of total memory

D.7 Experiment Validation Results

D.7.1 Ground Truth Validation

Manual evaluation of clustering results against expert judgment:

Table D.8: Clustering Accuracy vs Ground Truth

Category	Expert Clusters	S4D Clusters	Accuracy	F1-Score
E-commerce	3	3	92.3%	0.89
News/Media	2	3	87.5%	0.85
Educational	2	2	95.0%	0.93
Corporate	2	2	88.9%	0.87
Blog/Personal	1	2	83.3%	0.81
Government	1	1	100%	1.00
Overall	11	13	91.0%	0.89

D.7.2 Cross-Validation Results

5-fold cross-validation results across different website datasets:

Table D.9: Cross-Validation Performance

Fold	DBCV Score	Accuracy	Precision	Recall
1	0.72	89.2%	0.91	0.87
2	0.75	92.1%	0.93	0.91
3	0.69	87.5%	0.89	0.85
4	0.74	90.8%	0.92	0.89
5	0.71	88.9%	0.90	0.88
Mean	0.72	89.7%	0.91	0.88
Std	0.023	1.89%	0.016	0.023

D.8 Comparative Analysis

D.8.1 Baseline Comparisons

Comparison with traditional clustering approaches:

Table D.10: Algorithm Performance Comparison

Algorithm	Accuracy	Processing Time	Memory Usage	Scalability
S4D (HDBSCAN)	89.7%	12.8s	0.9GB	Excellent
K-Means	67.2%	3.2s	0.4GB	Good
Hierarchical	72.8%	45.6s	2.1GB	Poor
DBSCAN	78.4%	8.9s	0.7GB	Good
Spectral	71.5%	67.3s	3.2GB	Poor

D.8.2 Feature Importance Analysis

Analysis of which features contribute most to clustering quality:

Table D.11: Feature Importance Rankings

Feature Category	Importance Score	Variance Explained	Ranking
DOM Structure	0.34	28.7%	1
CSS Layout Properties	0.28	23.1%	2
Color Scheme	0.19	15.8%	3
Typography	0.12	9.9%	4
Interactive Elements	0.07	5.8%	5

D.9 Error Analysis

D.9.1 Failure Case Analysis

Analysis of URLs that failed to process correctly:

Table D.12: Failure Analysis

Failure Type	Count	Percentage	Common Causes
Timeout	12	42.9%	Heavy JavaScript, slow servers
DOM Parsing	8	28.6%	Malformed HTML, dynamic content
CSS Extraction	5	17.9%	Complex CSS rules, imports
Network Error	3	10.7%	Connection issues, blocked requests
Total	28	100%	Various technical issues

D.9.2 Quality Degradation Factors

Factors that negatively impact clustering quality:

- **Excessive DOM depth** (>10 levels): 15% quality reduction
- **Dynamic content loading**: 23% quality reduction
- **Minimal CSS styling**: 12% quality reduction
- **Mixed content types**: 8% quality reduction
- **Responsive design variations**: 6% quality reduction

D.10 Performance Optimization Results

D.10.1 Optimization Techniques Impact

D.11 Statistical Significance Tests

D.11.1 Hypothesis Testing Results

H1: S4D clustering is significantly better than random clustering

- **Test:** Paired t-test with $p < 0.001$

Table D.13: Performance Optimization Impact

Optimization	Before	After	Improvement
Browser instance reuse	25.6s	12.8s	50.0%
Vectorized operations	18.9s	9.4s	50.3%
DOM processing optimization	15.2s	8.9s	41.4%
Parallel processing	14.7s	8.9s	39.5%
Memory usage optimization	2.1GB	0.9GB	57.1%

- **Effect size:** 2.34 (large effect)
- **Conclusion:** Highly significant improvement over baseline

H2: Multi-level analysis improves clustering quality

- **Test:** One-way ANOVA with $F = 15.67$, $p < 0.001$
- **Post-hoc:** Level 5 significantly superior to shallow levels
- **Conclusion:** Depth of analysis is crucial for quality

H3: System optimizations significantly improve performance

- **Test:** Wilcoxon signed-rank test
- **W-statistic:** 1,247
- **p-value:** < 0.001
- **Conclusion:** Highly significant performance improvement with optimizations

D.12 Reproducibility Information

D.12.1 Experimental Configuration

System Configuration:

- **OS:** Ubuntu 20.04 LTS
- **Python:** 3.9.7
- **Chrome:** 96.0.4664.45
- **Memory:** 16GB RAM
- **CPU:** Intel i7-10700K

Random Seeds:

- **NumPy:** 42
- **Scikit-learn:** 42
- **HDBSCAN:** 42

D.12.2 Data Availability

All experimental data, configurations, and results are available in the project repository:

- **Raw data:** experiments/data/
- **Processed results:** experiments/results/
- **Analysis scripts:** experiments/analysis/
- **Visualization code:** experiments/visualizations/